

Self-Exciting Hawkes Jump Models

Introduction to the Paper

A key challenge in modelling oil prices is that large price movements do not arrive independently. In practice, a sudden crash tends to trigger further extreme moves shortly afterwards — a feature known as *jump clustering*. Standard jump-diffusion models handle jumps through a Poisson process with a *fixed* arrival rate, which cannot reproduce this behaviour. Gonzato and Sgarra (2020) address this by proposing a *self-exciting* jump-diffusion model where the jump intensity is stochastic and path-dependent: it rises after each jump and gradually decays back to a long-run background level. Combined with stochastic volatility and a stochastic convenience yield, the model is far more realistic than earlier factor models and is shown to improve both in-sample fit and out-of-sample futures forecasting for WTI crude oil data from 2008 to 2018.

What the Paper Introduced

The key building block is the *Hawkes process*, whose conditional intensity λ_t satisfies:

$$d\lambda_t = \beta(\lambda_\infty - \lambda_t) dt + \alpha dN_t, \quad (1)$$

where $\lambda_\infty > 0$ is the background intensity, $\beta > 0$ controls the speed of decay after a jump, and $\alpha > 0$ is the magnitude of the self-exciting spike caused by each new event dN_t . Every jump immediately raises λ_t by α , making the next jump more likely; the effect then fades exponentially, so calm periods are also possible. This reproduces *jump clustering* directly through the model structure. The log-price $x_t = \ln(S_t/S_0)$ follows:

$$dx_t = \left(\mu - \frac{1}{2}V_t - \lambda_t \mathbb{E}[e^{J_x} - 1] - \delta_t\right) dt + \sqrt{V_t} dW_t + dJ_{x,t}, \quad (2)$$

where V_t is a CIR-type stochastic variance that co-jumps with the price, and δ_t is an Ornstein–Uhlenbeck convenience yield. The affine structure of the system allows futures prices to be computed in closed form, greatly simplifying estimation.

Why Hawkes Processes Matter

In a standard Poisson model, the jump probability on any given day is independent of the past, which is clearly unrealistic. After a major oil market shock — a geopolitical event, a supply disruption — uncertainty stays elevated and further large moves remain more likely for days or weeks. Hawkes processes model this *financial contagion* directly: each jump raises the intensity, so consecutive extreme events are more probable. This makes them far better suited than constant-intensity models for capturing the turbulent, clustered dynamics that dominate oil markets during crises, and for producing realistic estimates of tail risk and derivative prices.

Bayesian Estimation and Sequential Monte Carlo

Estimation is complicated by the fact that V_t , δ_t , and λ_t are all *latent states* that cannot be observed directly. The authors use the *two-stage Sequential Monte Carlo (SMC) sampler*: a swarm of weighted random samples (*particles*) approximates the posterior of the parameters and latent states, being gradually tempered from the prior to the posterior via adaptive importance sampling and resampling. A first stage with $M_1 = 30$ state particles explores the parameter space

cheaply; a second stage with $M_2 = 500$ particles corrects the approximation. The method also delivers the marginal likelihood, enabling formal Bayes-factor model comparison — something that would be very difficult with classical maximum-likelihood approaches given the non-linear, non-Gaussian state-space structure.

Empirical Results

Applied to 2,767 daily WTI observations (spot plus eight futures contracts, 2008–2018), the results strongly support self-excitation. Log-returns exhibit kurtosis of 7.31 and fail the Jarque–Bera normality test. **Model I** (Hawkes dynamics) is compared to **Model II** (constant intensity, $\alpha = \beta = 0$): the log Bayes factor equals 9.13, constituting *decisive* evidence for Model I. The self-excitation parameter is estimated at $\hat{\alpha} = 135.19$ (SD 5.02), confirming that jump clustering is a genuine feature of the oil market. Out of sample, Model I also achieves lower Mean Absolute Error in futures pricing across all maturities.

Strengths of the Paper

The paper models jump dynamics in a flexible and realistic way, with the Hawkes intensity adapting automatically to market conditions. Stochastic volatility with co-jumps captures the leverage effect, and the affine structure keeps futures pricing analytically tractable. The model also has clear practical value: two optimal time-varying hedge ratios are derived, and the skewness-adjusted variance–skewness strategy outperforms pure minimum-variance hedging both in and out of sample.

Limitations of the Paper

The main drawbacks are computational. Running the two-stage SMC sampler is resource-intensive, and the large number of parameters increases calibration difficulty and the risk of overfitting. Results may also be somewhat sensitive to the choice of prior distributions, even though the authors report low prior sensitivity for most parameters. Overall, the Hawkes framework is considerably harder to implement than a standard constant-intensity model, which may limit its practical adoption.

Connection to Our Project

This paper is directly relevant because technology-heavy markets like QQQ also exhibit clustered extreme movements: crashes tend to arrive in waves rather than as isolated events. The Merton model assumes constant Poisson intensity and therefore underestimates the probability of consecutive large moves, while the Kou model improves jump asymmetry but keeps the intensity fixed. Hawkes-type models represent a natural frontier extension, allowing the intensity to respond dynamically to market shocks. Although implementing a full SMC-based self-exciting model is beyond the scope of this project, understanding how Hawkes processes extend traditional jump-diffusion frameworks provides important context for interpreting the results obtained with the Merton and Kou models and motivates the modelling choices made throughout the report.